STATA Tutorial 3B

Data Transformation

TIAN, Bing

March 11, 2025

Outline

Data Structure

Data Transformation
- append
- merge
- reshape
- collapse

Data Transformation

- Sometimes we want to make more "radical" changes to the dataset, rather than merely working on the variables. For example, we may want to:
  1. combine information from two datasets
  2. convert between long and wide data formats
  3. aggregate data to get group-level summary statistics
- Let's first see some common data types.

Common Data Structures

- cross-sectional data: data collection carried out at a single point in time on a statistical unit.
- pooled cross-sectional data: multiple cross-sectional data pooled together.
- time series data: a sequence of data points indexed in time order.
- panel data: a collection of quantities for multiple individuals, assembled over even time intervals and ordered chronologically.
- multilevel data: data for participants are organized at more than one level. The units of analysis are usually individuals (at a lower level) who are nested within contextual/aggregate units (at a higher level).
- paired data: datasets where every data point in one independent sample would be paired-uniquely-to a data point in another independent sample.

Cross-Sectional Data

```
. list, noobs clean

    id   var1   var2   var3
     1      .      .      .
     2      .      .      .
     3      .      .      .
     4      .      .      .
     5      .      .      .
     6      .      .      .
```

Pooled Cross-Sectional Data

```
. list, noobs clean

  sample   id   var1   var2   var3
    2022    1      .      .      .
    2022    2      .      .      .
    2022    3      .      .      .
    2023    4      .      .      .
    2023    5      .      .      .
    2023    6      .      .      .
```

Time Series Data

```
. list, noobs clean

    id   year   var1   var2   var3
     1   2016      .      .      .
     1   2017      .      .      .
     1   2018      .      .      .
     1   2019      .      .      .
     1   2020      .      .      .
     1   2021      .      .      .
```

Panel Data

```
. list, noobs clean

  id   year   var1   var2   var3
   1   2016      .      .      .
   1   2017      .      .      .
   1   2018      .      .      .
   2   2016      .      .      .
   2   2017      .      .      .
   2   2018      .      .      .
```

Multilevel Data

```
. list, noobs clean

    hid   pid   var1   var2   var3
      1     1      .      .      .
      1     2      .      .      .
      1     3      .      .      .
      2     1      .      .      .
      2     2      .      .      .
      2     3      .      .      .
```

Paired Data

```
. list, noobs clean

  hid   pid   sid   sinfo   var1   var2   var3
    1     1     2       .      .      .      .
    1     2     1       .      .      .      .
    1     3     .       .      .      .      .
    2     1     2       .      .      .      .
    2     2     1       .      .      .      .
    2     3     .       .      .      .      .
```

Outline

Data Structure

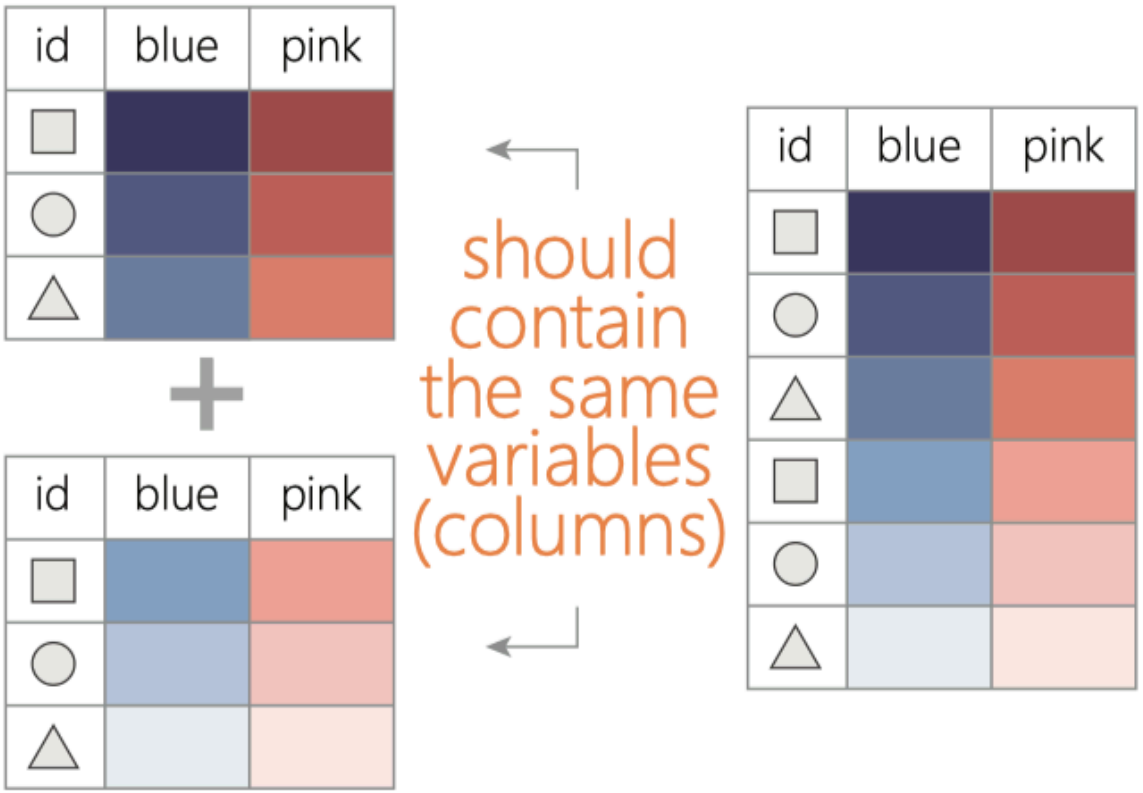
Data Transformation
- append
- merge
- reshape
- collapse

Outline

Data Structure

Data Transformation
- append
- merge
- reshape
- collapse

Combine Data: "append"



stata cheatsheet

Combine Data: "append"

```
. list, noobs clean

  dataset   id   male   age
        1    1      0    20
        1    2      0    18
        1    3      0    24
        1    4      1    32
        1    5      1    30
```

```
. list, noobs clean

  dataset   id   male   edu
        2    1      1     8
        2    7      1     5
        2    8      1     4
        2    9      1     3
        2   10      1     .
```

Combine Data: "append"

```
. use "$datadir/temp01", clear
. append using "$datadir/temp02"
. sort dataset id
. list, noobs clean
```

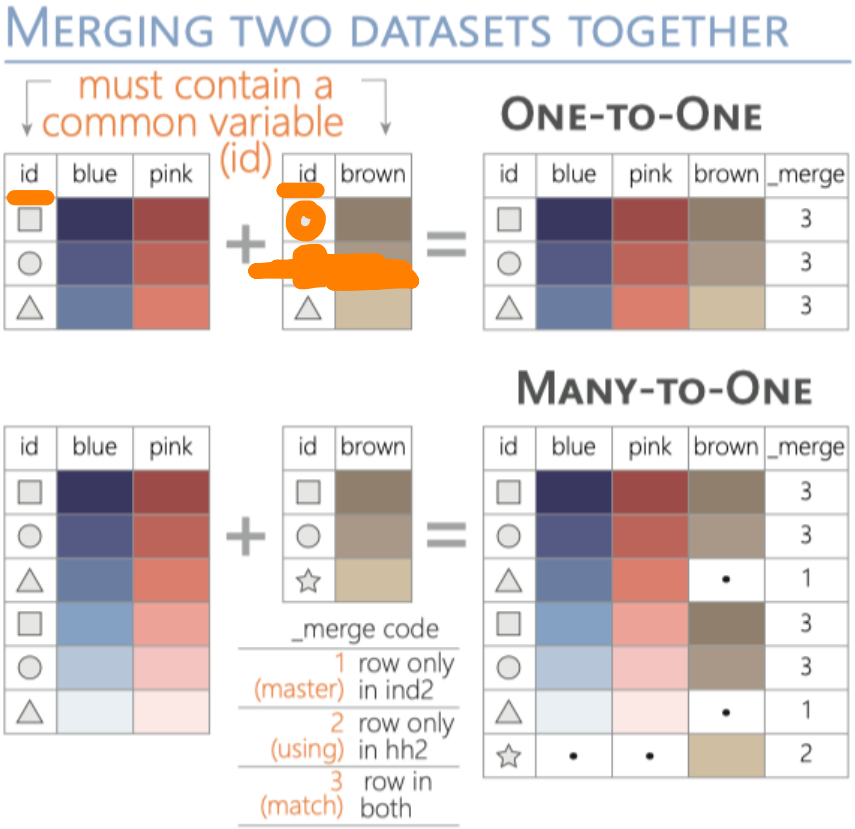
dataset	id	male	age	edu
1	1	0	20	.
1	2	0	18	.
1	3	0	24	.
1	4	1	32	.
1	5	1	30	.
2	1	1	.	8
2	7	1	.	5
2	8	1	.	4
2	9	1	.	3
2	10	1	.	.

Outline

Data Structure

Data Transformation
- append
- merge
- reshape
- collapse

Combine Data: "merge"




```
webuse ind_age.dta, clear
save ind_age.dta, replace
webuse ind_ag.dta, clear
```

merge 1:1 id **using** "ind_age.dta"

one-to-one merge of "ind_age.dta" into the loaded dataset and create variable "_merge" to track the origin

```
webuse hh2.dta, clear
save hh2.dta, replace
webuse ind2.dta, clear
```

merge m:1 hid **using** "hh2.dta"

many-to-one merge of "hh2.dta" into the loaded dataset and create variable "_merge" to track the origin

stata cheatsheet

Combine Data: "merge", 1:1

```
. use "$datadir/temp01", clear
. merge 1:1 id using "$datadir/temp02" //merge by id
    Result                           # of obs.
    -----------------------------------------
    not matched                             8
        from master                         4  (_merge==1)
        from using                          4  (_merge==2)
    matched                                 1  (_merge==3)
    -----------------------------------------

. sort id
. list, noobs clean
    dataset   id   male   age   edu            _merge
          1    1      0    20     8       matched (3)
          1    2      0    18     .   master only (1)
          1    3      0    24     .   master only (1)
          1    4      1    32     .   master only (1)
          1    5      1    30     .   master only (1)
          2    7      1     .     5    using only (2)
          2    8      1     .     4    using only (2)
          2    9      1     .     3    using only (2)
          2   10      1     .     .    using only (2)
. *for matched cases, taking values from master data
```

Combine Data: "merge", m:1

```
. list in 1/10, noobs clean

    dataset    id    male    age
          1     1       0     18
          1     1       0     19
          1     1       0     20
          1     2       0     18
          1     3       0     18
          1     3       0     19
          1     3       0     20
          1     3       0     21
          1     3       0     22
          1     3       0     23
```

```
. list, noobs clean

    dataset    id    male    edu
          2     1       1      8
          2     7       1      5
          2     8       1      4
          2     9       1      3
          2    10       1      .
```

Combine Data: "merge", m:1

```
. use "$datadir/temp03", clear
. merge m:1 id using "$datadir/temp02", keepusing(edu)
    Result                           # of obs.
    -----------------------------------------
    not matched                             40
        from master                         36  (_merge==1)
        from using                           4  (_merge==2)
    matched                                  3  (_merge==3)
    -----------------------------------------

. list in 1/10, sepby(id) noobs clean
    dataset   id   male   age   edu            _merge
          1    1      0    18     8       matched (3)
          1    1      0    19     8       matched (3)
          1    1      0    20     8       matched (3)
          1    2      0    18     .   master only (1)
          1    3      0    18     .   master only (1)
          1    3      0    19     .   master only (1)
          1    3      0    20     .   master only (1)
          1    3      0    21     .   master only (1)
          1    3      0    22     .   master only (1)
          1    3      0    23     .   master only (1)
```

Outline

Data Structure

Data Transformation
- append
- merge
- reshape
- collapse

Reshape Data: "reshape"

```
webuse set https://github.com/GeoCenter/StataTraining/raw/master/Day2/Data
webuse "coffeeMaize.dta"                    load demo dataset
```

Melt data (wide → long)

reshape variables starting with coffee and maize — unique id variable (key) — create new variable that captures the info in the column names

```
reshape long coffee@ maize@, i(country) j(year)
```

new variable

convert a wide dataset to long

Wide

country	coffee 2011	coffee 2012	maize 2011	maize 2012
Malawi				
Rwanda				
Uganda				

melt →

← **cast**

Long (Tidy)

country	year	coffee	maize
Malawi	2011		
Malawi	2012		
Rwanda	2011		
Rwanda	2012		
Uganda	2011		
Uganda	2012		

Tidy Datasets have each observation in its own row and each variable in its own column.

When datasets are tidy, they have a consistent, standard format that is easier to manipulate and analyze.

Cast data (long → wide)

create new variables named coffee2011, maize2012... — what will be unique id variable (key) — create new variables with the year added to the column name

```
reshape wide coffee maize, i(country) j(year)
```

convert a long dataset to wide

stata cheatsheet

Reshape Data: "reshape wide"

Reshape from long to wide

```
. reshape wide edu age, i(id) j(year)
(note: j = 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015)
Data                               long   ->   wide
-----------------------------------------------------------------------------
Number of obs.                       43   ->      9
Number of variables                   5   ->     32
j variable (15 values)             year   ->   (dropped)
xij variables:
                                    edu   ->   edu2001 edu2002 ... edu2015
                                    age   ->   age2001 age2002 ... age2015
-----------------------------------------------------------------------------

.         //don´t need list constant varialbes, e.g. "male"
. list id male edu2001 edu2002 age2001 age2002, noobs clean
```

id	male	edu2001	edu2002	age2001	age2002
1	0	8	8	18	19
2	0	.	.	18	.
3	0	.	.	18	19
4	1	.	.	18	19
5	1	.	.	18	19
7	.	5	.	.	.
8	.	4	.	.	.
9	.	3	.	.	.
10	.	.	.	.	.

Reshape Data: "reshape long"

Reshape from wide to long

```
. reshape long edu age, i(id) j(year)
(note: j = 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015)
Data                               wide   ->   long
-----------------------------------------------------------------------------
Number of obs.                        9   ->     135
Number of variables                  32   ->       5
j variable (15 values)                    ->   year
xij variables:
            edu2001 edu2002 ... edu2015   ->   edu
            age2001 age2002 ... age2015   ->   age
-----------------------------------------------------------------------------
.         //don´t need list constant varialbes, e.g. "male"
. drop if id==id[_n-1] & edu==. & age==.
(92 observations deleted)
.         //drop empty observations
. list in 1/10, noobs clean
    id   year   age   edu   male
     1   2001    18     8      0
     1   2002    19     8      0
     1   2003    20     8      0
     2   2001    18     .      0
     3   2001    18     .      0
     3   2002    19     .      0
     3   2003    20     .      0
     3   2004    21            0
```

Outline

Data Structure

Data Transformation
- append
- merge
- reshape
- collapse

Aggregate Data: "collapse"

```
. preserve //preserve current memory
. collapse (mean) age, by(id)
. list, noobs clean
    id   age
     1    19
     2    18
     3    21
     4    25
     5    24
     7     .
     8     .
     9     .
    10     .
. restore //restore to the state when we "preserve"
```